

# SENTINEL AI - ASSET DIAGNOSTIC REPORT

Automated Predictive Maintenance & Prognostics Engine

## 1. System Metadata

Timestamp: 2026-08-15 12:12:29

Target Asset: Rexnord ZA-2115 Bearing (IMS Test 3)

## 2. Operational Health Status

CRITICAL METRIC STATUS: SEVERE (30.57% Health Score)

## 3. Key Telemetry Feature Dimensions

| Feature Metric                   | Extracted Value | Baseline Reference |
|----------------------------------|-----------------|--------------------|
| RMS_mean (Continuous Energy)     | 0.3169 g        | < 0.0800 g         |
| RMS_max (Peak Shock Energy)      | 0.5900 g        | < 0.1200 g         |
| Kurtosis_mean (Signal Spikiness) | 3.2536          | < 1.0000           |
| PeakFreq_mean (Dominant Freq)    | 2243 Hz         | 1000 Hz            |
| PeakMag_max (FFT Peak Magnitude) | 1635.13         | < 400.00           |

## 4. Predictive Prognostics (Linear Trend Forecast)

CRITICAL SHUTDOWN ALERT: Sentinel AI has flagged an extreme operational hazard.

- Current Health Score: 30.57% (SEVERE)
- Notice: Structural failure is imminent. The asset has breached critical safety tolerances.

Action Required: Emergency structural intervention required immediately. Halt machine operations.